

Summary

Staff-level engineer shipping production agentic systems in financial services. Delivered a conversational AI portfolio assistant, a replayable workflow engine, and LLM evaluation pipelines. 10+ years in software dev (data, ML, AI).

Experience

Omega Point Research	<i>Senior Python Software Engineer</i>	August 2023 - Present
▷ Lead software engineer for AI products: architected conversational agents, agentic workflows, and LLM evaluation systems ▷ Built data pipelines using Spark and Databricks for risk model calculations; conduct technical interviews for eng. candidates ▷ Drove team adoption of agent-first development (voice dictation + AI coding tools) via demos, presentations, and coaching ▷ Trusted by CTO to float across teams and tackle high-priority problems; mentored intern through LLM research project		
ZAIS Group LLC	<i>Senior Software Architect</i>	April 2021 - August 2023
▷ Led internal tooling dev for a portfolio management firm with \$5.4B AUM; built data applications, analytics, and reporting ▷ Designed Azure cloud architecture; Airflow pipelines for multi-vendor position data synthesis; built portfolio analysis tools ▷ Collaborated with executives on product roadmap; self-taught structured credit domain to make informed technical decisions		
Black River Systems Co.	<i>Software Engineer</i>	May 2015 - April 2021
▷ Developed analytical software and ML solutions for government agencies; held Secret clearance for classified project work ▷ Authored two successful contract proposals (\$1.1M total); recognized for starting and leading a technical talk series		
Independent Consultant	<i>Data Science / Software Engineering</i>	2016, 2020, 2023
▷ Part-time consulting for financial clients (ZAIS, Wells Fargo, DBRS); PySpark pipelines, financial modeling, data cleaning		

Selected Projects

AI Teammate for Portfolio Analysis	Omega Point Research
▷ Owned Python service for AI chat: users ask portfolio questions, get data tables and charts—shipped with product launch ▷ Authored architecture after analyzing options; led cross-team discovery (AI, App, Platform); locked in API contracts early ▷ Built repository abstraction with dependency injection; added tracing for debugging; enabled AI team to test agents locally	
Plan-Execute Workflow Framework	Omega Point Research
▷ Designed the workflow framework that runs multi-step agent plans (query, transform, visualize) using replayable references ▷ Initial version rejected by team; built working examples, iterated on prototype, earned buy-in—pattern now in production ▷ Led terminology standardization via literature review; consolidated 2 competing approaches into a single unified framework	
AI Evaluation & Observability Pipeline	Omega Point Research
▷ Built LLM-as-Judge framework with checklist validation; transformed team from ad-hoc prompt tuning to regression testing ▷ Led vendor evaluation and Arize integration; enabled live evals and full reasoning traces for debugging production failures	
Index-as-a-Security Platform	Omega Point Research
▷ Built data pipelines mapping index constituents to risk models and computing factor exposures; processes 1B+ data points ▷ Set up vendor integrations and a monitoring dashboard; built Go-Python bridge for PySpark; achieved 60x write speedup ▷ Debugged C extension core dumps using GDB + LLM (early 2024); contributed 2 bug fixes upstream to gopy open source	
Highlight Helper	Side Project
▷ Built app that extracts text from photos using natural language queries; Vision API + LLM returns only requested content	
Financial Model Engineering	ZAIS
▷ Achieved 3x Monte Carlo runtime via profiling and Numba; prototyped 250x speedup with 500+ parallel AWS λ functions ▷ Loan implied ratings: translated SME’s Excel model to Python CLI + Teams Bot; reformulated algo as quadratic program ▷ Position reconciliation: matched transactions across vendors to derive cash balances; drew manager praise post-departure	
Edge-Based Threat Detection System	Black River Systems
▷ Led a 3-engineer team building deep learning system for real-time signal classification on edge hardware; Keras, Ray, Docker	

Skills

AI/ML: Production agentic systems, LLM evaluation (LLM-as-Judge), prompt engineering, RAG, DSPy, Claude/OpenAI, Arize
AI Dev Tools: Claude Code, Cursor, agent orchestration, custom skills/prompts, MCP (Model Context Protocol)
Languages: Python, SQL, Go, Scala, Java, $\LaTeX$ **Data:** PySpark, Databricks, Airflow, Delta Lake, medallion architecture
Cloud/Infra: AWS (Lambda, S3), Azure, Docker, FastAPI, PostgreSQL

Education

University of Minnesota Duluth	<i>M.S. Applied Mathematics, CS Minor</i>	May 2015
Bethany Lutheran College	<i>B.A. Mathematics, Magna Cum Laude</i>	May 2013